

School of computer Science and engineering

B. Tech CSE- AIML

Time Series Analysis and Forecasting Techniques (23BSCSMA61)

Experiential learning activity on a
“Statistical Analysis and Time Series Forecasting of Retail Sales
Data”

Submitted to

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1.Problem Statement and Significance:

Problem Statement

Retail companies must understand sales patterns to manage inventory, pricing strategies, and supply chain operations effectively. In this project, we analyze the historical sales data of IKEA India to identify time-dependent patterns in sales performance and forecast future sales.

The dataset contains information about orders including order date, product category, units sold, price, discount percentage, and total sales. Since the dataset records sales transactions across different dates, it can be treated as time series data.

The primary objective of this study is to perform time series analysis on the sales data and develop forecasting models to predict future sales trends.

Time Interval

The time variable used in this study is Order_Date, which represents the date when a transaction occurred. The data will be aggregated into monthly sales values to observe long-term trends and seasonal patterns.

Objective of the Analysis

The objectives of this project are:

- To analyze the trend in IKEA India sales over time
- To identify seasonal patterns or fluctuations in sales
- To study the relationship between time and total sales
- To apply time series forecasting models
- To predict future sales values

Significance of the Study

Sales forecasting is very important for retail businesses because it helps companies:

- Plan inventory levels efficiently
- Optimize supply chain operations
- Improve pricing and discount strategies
- Make informed business decisions

By applying time series forecasting techniques to the IKEA sales dataset, businesses can better understand sales behaviour and prepare for future demand

2. Data Collection

Data Source

The dataset used in this project is a secondary dataset titled “IKEA India Large Sales Dataset.” The dataset contains historical transaction data for IKEA product sales across different states and cities in India.

The dataset includes 2000 records and 12 attributes, providing detailed information about sales transactions.

Click here: [DATASET](#)

DatasetAttributes

The dataset contains the following important attributes:

Attribute	Description
Order_ID	Unique identifier for each order
Order_Date	Date of the transaction
State	State where the purchase occurred
City	City where the purchase occurred
Product_Category	Category of IKEA product
Units_Sold	Number of units sold
Unit_Price	Price per unit
Discount_Percent	Discount applied
Total_Sales	Total sales value
Profit	Profit from the sale
Payment_Method	Mode of payment
Customer_Type	Member or normal customer

Selected Variables forAnalysis

For the time series analysis, the following variables are selected:

- Order_Date → Time variable
- Total_Sales → Main forecasting variable
- Units_Sold → Supporting analysis
- Profit → Additional trend analysis

Data Preparation

Before performing time series analysis, the following preprocessing steps are performed:

1. Convert Order_Date into a date format.
2. Sort the dataset based on date.
3. Aggregate sales values by month to create a time series.
4. Handle missing values or inconsistencies if present.

These steps ensure that the dataset is properly structured for time series analysis and forecasting.

3. Time Series Analysis Methods

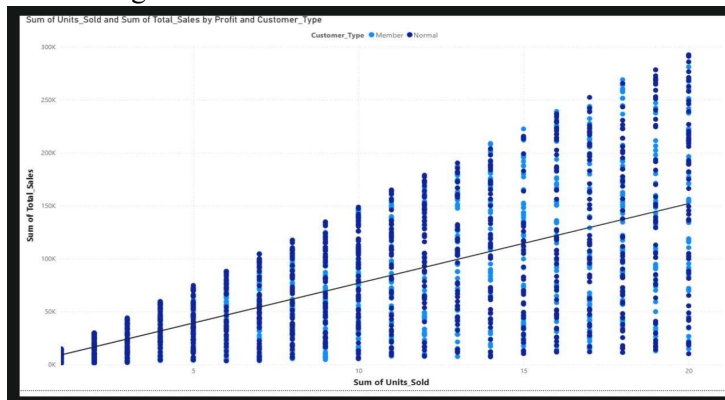
Time series data consists of several components such as trend, seasonality, cyclical variation, and irregular variation. In this study, various analytical techniques are used to identify these components.

Trend Analysis

Trend represents the long-term movement of data over time. In this project, trend analysis helps determine whether IKEA sales are increasing, decreasing, or remaining stable over time.

Trend can be analyzed using:

- Time series plots
- Moving averages
- Regression models



Seasonal Analysis

Seasonality refers to repeating patterns that occur at regular intervals. Retail sales often show seasonal fluctuations due to festivals, holidays, or promotional events.

By analyzing monthly sales data, we can identify whether sales increase during specific periods.

Moving Average Method

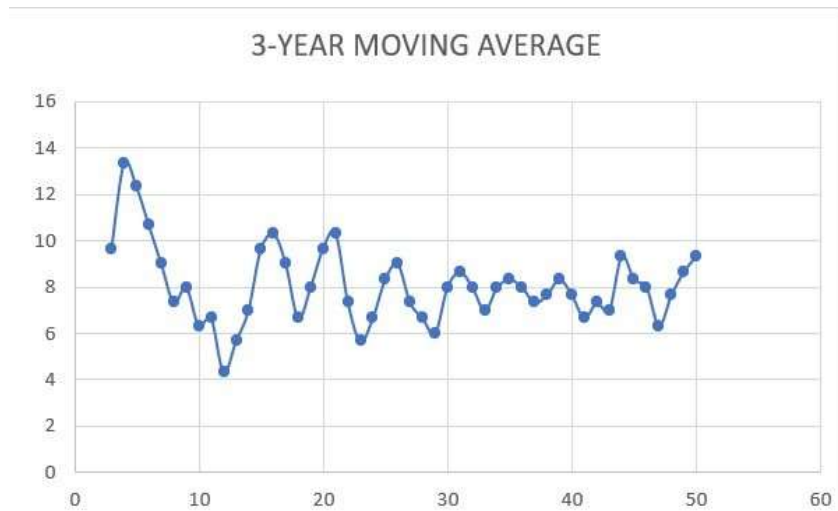
The moving average method is used to smooth fluctuations in the data and highlight the underlying trend.

For example:

3-month moving average can be calculated as:

$$MA = \frac{Sales_{t-1} + Sales_t + Sales_{t+1}}{3}$$

This helps reduce irregular fluctuations and reveal the general direction of sales.

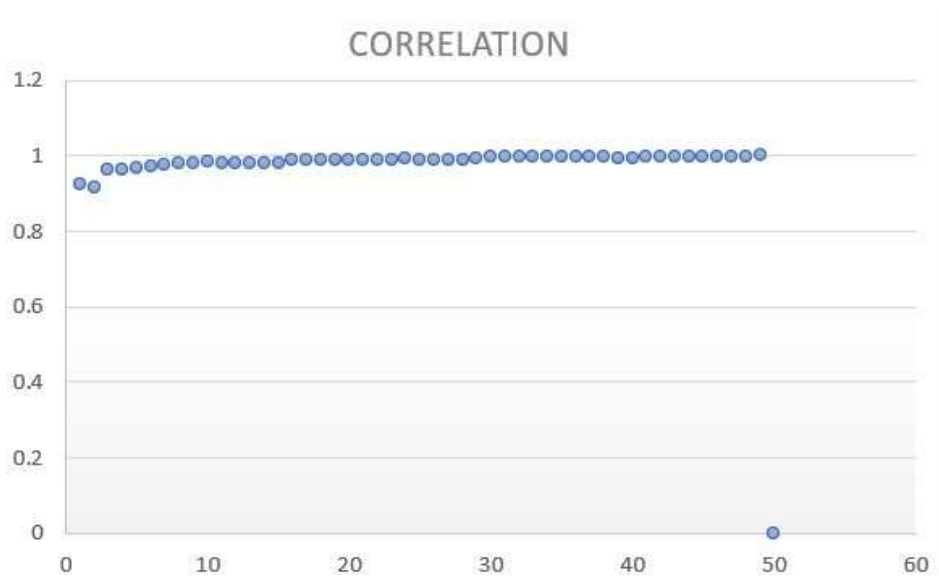


Correlation Analysis

Correlation analysis can be used to examine the relationship between:

- Units Sold
- Total Sales
- Profit

Lagged correlation may also be used to study relationships between previous and future values.



Data Transformation

To make the time series suitable for forecasting models, certain transformations may be applied.

These include:

- De-trending – Removing the long-term trend component
- De-seasonalization – Removing seasonal effects

- Non Stationarity testing – Ensuring statistical properties remain Fluctuat in gover time.

ARIMA works good for non stationary data

Hypothesis Formulation

Hypothesis testing is a statistical method used to determine whether there is enough evidence to support a specific claim about a dataset. In this study, hypothesis testing is applied to examine the relationship between Total Sales and Profit. Statistical techniques such as the Z-test and T-test are used to validate the significance of this relationship.

- Null Hypothesis (H_0):
There is no significant relationship between Total Sales and Profit.
- Alternative Hypothesis (H_1):
There is a significant relationship between Total Sales and Profit.

The correlation coefficient (r) is used to measure the strength and direction of the relationship between Total Sales and Profit.

Formula:

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \cdot \sum(Y - \bar{Y})^2}}$$

Where:

- XXX = Total Sales
- YYY = Profit
- $\bar{X}, \bar{Y}$ = Mean values

Decision Rule:

- If p-value $< 0.05 \rightarrow$ Reject H_0
- If p-value $> 0.05 \rightarrow$ Fail to reject H_0

The Z-test is applied when the sample size is large (≥ 30) and population variance is known, to test the mean of the dataset.

Formula:

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

Where:

- $\bar{X}$ = sample mean
- μ = population mean
- σ = standard deviation
- n = sample size

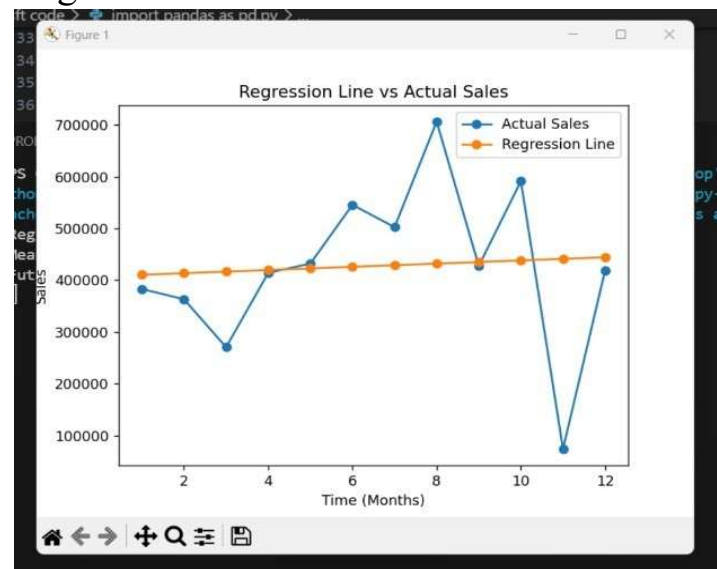
Explanation of Results

- The correlation coefficient (r) indicates the strength and direction of the relationship:
 - Positive value $\rightarrow$ Sales and Profit increase together
 - Negative value $\rightarrow$ Inverse relationship
- ◻ The p-value helps in decision-making:
 - $p < 0.05 \rightarrow$ Strong evidence against H_0
 - $p > 0.05 \rightarrow$ Insufficient evidence to reject H_0

Interpretation

- A high positive correlation implies that higher sales lead to higher profit.
- If the null hypothesis is rejected, it confirms that a significant relationship exists between Total Sales and Profit.

Regression line:

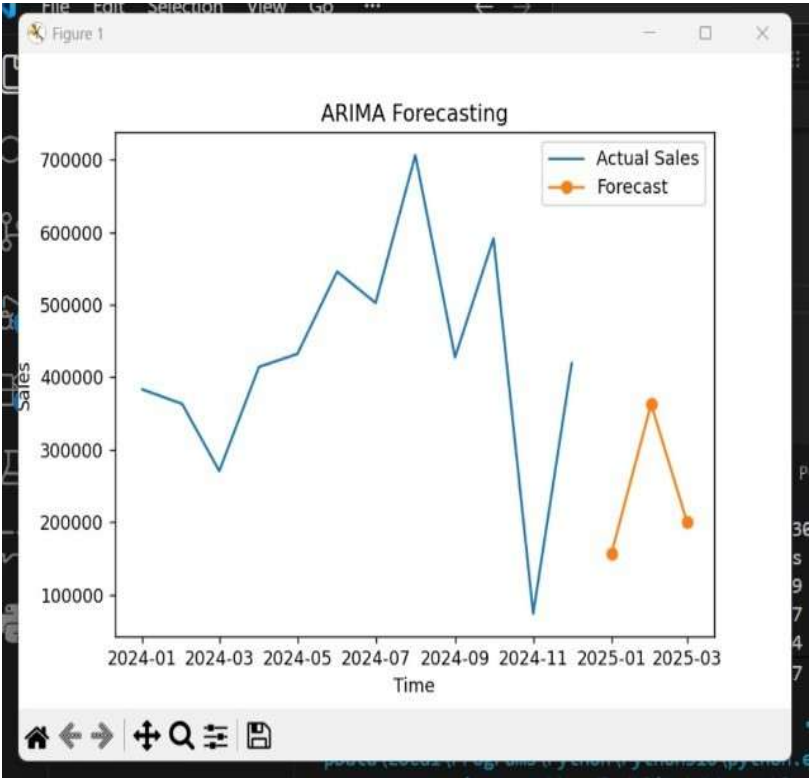


Output:

```
PS C:\Users\Dell\Downloads\workshop> c::; cd 'c:\Users\Dell\Downloads\workshop'; & 'C:\Users\Dell\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\Dell\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\l
xtensions (Ctrl+Shift+X) - 3 require restart ell\Downloads\workshop\tsft code\import pandas as pd.py'
Regression Line: y = 407388.42 + 3083.56t
Mean Squared Error: 23251552937.13335
Future Predictions: [447474.74242424 450558.30536131 453641.86829837]
```


The graph compares actual sales with a regression line over 12 months. The actual sales (blue line) show large fluctuations, with noticeable peaks and drops, indicating inconsistency. In contrast, the regression line (orange) shows a steady upward trend, suggesting that overall sales are gradually increasing despite short-term variations.

ARIMA model:



OUTPUT:

```
plt.show()

... /usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:966: UserWarning: Non-stationary starting autoregressive parameters found. Using zeros as starti
warn('Non-stationary starting autoregressive parameters')
/usr/local/lib/python3.12/dist-packages/statsmodels/tsa/statespace/sarimax.py:978: UserWarning: Non-invertible starting MA parameters found. Using zeros as starting parameter
warn('Non-invertible starting MA parameters found.')
ARIMA Forecast:
2025-01-01    156264.102326
2025-02-01    362825.175306
2025-03-01    200382.593185
Freq: MS, Name: predicted_mean, dtype: float64

Future Predictions: [447474.74242424 450558.30536131 453641.86829837]
  Month  Total_Sales  t  Predicted_Sales
0  2024-01      383064   1    410471.987179
1  2024-02      363164   2    413555.550117
2  2024-03      270521   3    416639.113054 ...
11 2024-12      418926  12    444391.179487
PS C:\Users\Dell\Downloads\workshop> ^C
PS C:\Users\Dell\Downloads\workshop> c:: cd 'c:\Users\Dell\Downloads\workshop'; & 'C:\Users\Dell\AppData\Local\Programs\Python\Python310\python.exe' 'c:\Users\Dell\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '59313' '--' 'c:\Users\Dell\Downloads\workshop\tsft code\import pandas as pd.py'
Regression Line: y = 407388.42 + 3083.56t
Mean Squared Error: 23251552937.13335
Future Predictions: [447474.74242424 450558.30536131 453641.86829837]
```

Conclusion:

The time series analysis of the sales dataset shows that there is an overall increasing trend in sales over time. Using the regression model, we observed a steady growth pattern, but it could not fully capture fluctuations in the data.

To improve prediction accuracy, the ARIMA model was applied. The ARIMA model considers past values and patterns, making it more suitable for time series forecasting. The forecast results indicate a continued upward trend in future sales.

Overall, the study demonstrates that time series techniques are effective in understanding sales behavior and predicting future trends, which can help businesses make better decisions regarding inventory and planning.

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